

Relevance-Based Prediction: Factor Selection for Saugus Schools

Four outcomes: MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, Education Budget Share

One RBP run per outcome on ALL candidate variables — no feature pruning (faithful to Kritzman)

Exhibit 5 importance is descriptive: positive = helps, ≈ 0 = benign / diversified away (fn. 12), < 0 = harmful

Saugus analyzed as the prediction task; most/least relevant towns identified per Exhibit 4

Leave-one-out LOO r validates the predictions across all MA districts

Model Summary (all candidates used — no pruning)			
MCAS Grades 3-8	Dropout Rate	MCAS Grade 10 (ELA)	Education Budget Share
Predicted on 19 candidates	Predicted on 19 candidates	Predicted on 19 candidates	Predicted on 19 candidates
Positive importance: 18 of 19	Positive importance: 18 of 19	Positive importance: 18 of 19	Positive importance: 18 of 19
LOO r = +0.902	LOO r = +0.838	LOO r = +0.900	LOO r = +0.801

How to Read This Report — Three Tiers of Factors

This report matches Saugus to demographically-similar towns using Tier 3, then ranks the Tier 1 & 2 levers Saugus can actually change. Structural traits are the backdrop for finding peers — never a recommendation.

TIER 1 — Directly votable (Town Meeting / ballot)

What the town changes by a vote.

- Education's share of the municipal budget
- Residential tax rate · Proposition 2½ override · debt exclusion
- Spending above the state-required school minimum (Ch. 70)
- Reserves (free cash, stabilization) · OPEB funding · capital projects

TIER 2 — Policy / management (administration decides)

What the school department & town administration run day-to-day (funded by Tier 1).

- Chronic absenteeism — attendance & re-engagement programs
- Teacher staffing levels and pay
- How the school dollar splits between classroom and overhead
- Health-insurance plan design / GIC membership

TIER 3 — Structural (NOT changeable by vote)

What the community IS — used ONLY to find Saugus's true peer towns.

- Household income · property wealth · adult education
- Poverty · English-learner · special-education shares
- Housing tenure · district enrollment · regional economy
- → Defines the peer group; never shown as a 'lever' in this report.

Combined Model Summary: All Four Outcomes

RBP factor selection results across MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, and Education Budget Share

Model	Predicted	Actual	Gap	Direction	LOO r	Features
MCAS Grades 3–8	23.6%	29.0%	+5.4pp	over	+0.902	19
Dropout Rate	4.1%	2.9%	-1.2pp	over	+0.838	19
MCAS Grade 10 (ELA)	34.9%	47.0%	+12.1pp	over	+0.900	19
Education Budget Share	40.9%	32.3%	-8.6pp	under	+0.801	19

Actionable Lever Cross-Reference — Which Levers Matter Across Outcomes

Actionable levers only (structural traits are matching-only). A lever that matters in multiple outcomes is a robust target.

Feature	# Models	MCAS 3–8	Dropout	MCAS 10	Ed Budget
chronic_absenteeism_pct	4	+0.541	+0.908	+0.622	-0.026
nss_per_eqv	4	+0.306	+0.282	+0.442	+0.270
teacher_share_of_spend	4	+0.378	+0.327	+0.361	+0.175
teachers_per_lowincome	4	+0.375	+0.335	+0.352	+0.127
teachers_per_100_students	4	+0.359	+0.387	+0.346	+0.079
res_tax_rate	4	+0.322	+0.325	+0.349	+0.149
spend_vs_required_nss	4	+0.377	+0.314	+0.374	+0.064
health_ins_per_capita	4	+0.300	+0.335	+0.340	+0.090
nss_per_pupil	4	+0.326	+0.233	+0.357	+0.047
ed_budget_share	3	-0.116	-0.218	-0.163	—
fixed_costs_pct	1	—	—	—	+0.393

Green rows = feature appears in 3+ models. Cell colour: green = positive importance, red = negative importance, white = not in that model (—).

What This Means: MCAS Grades 3-8

Predicted 23.6pp · Actual 29.0pp · Gap +5.4pp (higher is better) · Saugus ranks 54 of 219 MA districts on this measure

The Bottom Line

Saugus's MCAS grades 3–8 proficiency (ELA + Math, meeting/exceeding) is 29.0%, 5.4pp better than its demographic prediction of 23.6% — ranking 54 of 219 MA districts, in the top 25%. A district with Saugus's income, poverty, and enrollment profile is predicted to land near this level.

Top Actionable Levers (what Saugus can change)

Highest-importance actionable factors, Saugus vs. statewide median. Structural traits match peers silently and are not shown here.

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.54
Teacher pay share of school spending	0.349	0.368	+0.38
Spending vs Ch70 required minimum (×)	1.03	1.36	+0.38
Teachers per 100 ÷ low-income %	0.126	0.277	+0.37
Teachers per 100 students	5.87	8.07	+0.36

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	19.1%	-12.1pp
Teacher pay share of school spending	0.349	0.359	+0.010
Spending vs Ch70 required minimum (×)	1.03	1.54	+0.504
Teachers per 100 ÷ low-income %	0.126	0.213	+0.087
Teachers per 100 students	5.87	8.81	+2.93

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

Saugus does better here than its demographic profile predicts. The strongest RBP driver of Saugus's predicted peer group is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the over-performing districts is students chronically absent (10%+) (-12.1pp, peer median – Saugus). (Driver ranking is close enough at the top that the #1 factor is not fully seed-stable — read the top group, not the exact order.) The drivers below are where Saugus differs most from the towns that beat their own predictions — the most plausible places to move this result.

What Overachievers Do Differently: MCAS Grades 3-8

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 10 shown

Feature	Saugus	Amherst	Rowe	Brimfield	Imp
chronic_absenteeism_pct	31.2	16.7	26.2	20.3	+0.541
teacher_share_of_spend	0.35	0.31	0.33	0.46	+0.378
spend_vs_required_nss	1.03	1.76	1.91	1.36	+0.377
teachers_per_lowincome	0.13	0.28	0.20	0.29	+0.375
teachers_per_100_student	5.87	9.80	10.6	9.23	+0.359
nss_per_pupil	17,369	31,268	34,298	20,046	+0.326
res_tax_rate	10.7	18.5	5.01	14.6	+0.322
nss_per_eqv	0.07	0.35	0.03	0.12	+0.306
health_ins_per_capita	588.1	4.36	1,671	81.4	+0.300
ed_budget_share	32.3	52.2	29.9	59.8	-0.116

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	29.0	23.6	+5.4pp
Amherst	49.5	7.0	+42.5pp
Rowe	36.0	10.7	+25.3pp
Brimfield	66.5	44.3	+22.2pp
Eastham	55.0	34.5	+20.5pp
Oak Bluffs	44.0	23.8	+20.2pp
Brookfield	51.0	34.3	+16.7pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Dropout Rate

Predicted 4.1pp · Actual 2.9pp · Gap -1.2pp (lower is better) · Saugus ranks 27 of 168 MA districts on this measure

The Bottom Line

Saugus's dropout rate is 2.9%, 1.2pp better than its demographic prediction of 4.1% — ranking 27 of 168 MA districts, in the top 25%. Dropout reflects re-engagement and attendance recovery as much as demographics.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	24.3%	-6.9pp
Teachers per 100 students	5.87	7.46	+1.59
Teachers per 100 ÷ low-income %	0.126	0.133	+0.007
Health insurance \$ per resident	\$588	\$632	+44
Teacher pay share of school spending	0.349	0.366	+0.017

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

Top Actionable Levers (what Saugus can change)

Highest-importance actionable factors, Saugus vs. statewide median. Structural traits match peers silently and are not shown here.

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.91
Teachers per 100 students	5.87	8.07	+0.39
Teachers per 100 ÷ low-income %	0.126	0.277	+0.33
Health insurance \$ per resident	\$588	\$453	+0.33
Teacher pay share of school spending	0.349	0.368	+0.33

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

The Takeaway

Saugus does better here than its demographic profile predicts. The strongest RBP driver of Saugus's predicted peer group is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the over-performing districts is health insurance \$ per resident (+44, peer median – Saugus). (Driver ranking is close enough at the top that the #1 factor is not fully seed-stable — read the top group, not the exact order.) Chronic absenteeism is the factor to watch — the main risk to this result if it climbs.

What Overachievers Do Differently: Dropout Rate

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by [importance], top 10 shown

Feature	Saugus	Marlborou	Clinton	Worcester	Imp
chronic_absenteeism_pct	31.2	23.5	7.70	28.3	+0.908
teachers_per_100_student	5.87	7.36	7.31	7.40	+0.387
teachers_per_lowincome	0.13	0.13	0.14	0.10	+0.335
health_ins_per_capita	588.1	794.6	362.4	700.1	+0.335
teacher_share_of_spend	0.35	0.37	0.39	0.38	+0.327
res_tax_rate	10.7	10.2	13.1	13.8	+0.325
spend_vs_required_nss	1.03	1.11	1.08	1.10	+0.314
nss_per_eqv	0.07	0.09	0.12	0.20	+0.282
nss_per_pupil	17,369	20,169	18,512	22,244	+0.233
ed_budget_share	32.3	43.0	49.6	45.8	-0.218

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	2.9	4.1	-1.2pp
Marlborough	1.4	4.5	+3.1pp better
Clinton	1.1	4.2	+3.1pp better
Worcester	2.1	5.0	+2.9pp better
Palmer	0.9	3.4	+2.5pp better
Norwood	0.1	2.5	+2.4pp better
Fitchburg	2.1	4.5	+2.4pp better

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: MCAS Grade 10 (ELA)

Predicted 34.9pp · Actual 47.0pp · Gap +12.1pp (higher is better) · Saugus ranks 10 of 168 MA districts on this measure

The Bottom Line

Saugus's grade 10 ELA proficiency is 47.0%, 12.1pp better than its demographic prediction of 34.9% — ranking 10 of 168 MA districts, in the top 10% of MA districts. Note: the grade 3–8 academic base is an outcome, not a lever, so it is excluded from this model — part of the over-performance shown here reflects that omitted prior strength rather than a grade-10 effect.

Top Actionable Levers (what Saugus can change)

Highest-importance actionable factors, Saugus vs. statewide median. Structural traits match peers silently and are not shown here.

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.62
School spending ÷ property wealth/capita	0.068	0.082	+0.44
Spending vs Ch70 required minimum (×)	1.03	1.36	+0.37
Teacher pay share of school spending	0.349	0.368	+0.36
Net school spending per pupil	\$17,369	\$20,865	+0.36

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	18.9%	-12.3pp
School spending ÷ property wealth/cap	0.068	0.090	+0.023
Spending vs Ch70 required minimum (÷)	1.03	1.40	+0.369
Teacher pay share of school spending	0.349	0.363	+0.014
Net school spending per pupil	\$17,369	\$24,272	+6,904

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

Saugus does better here than its demographic profile predicts. The strongest RBP driver of Saugus's predicted peer group is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the over-performing districts is net school spending per pupil (+6,904, peer median – Saugus). (Driver ranking is close enough at the top that the #1 factor is not fully seed-stable — read the top group, not the exact order.) Among the actionable drivers, chronic absenteeism and teacher staffing are the most visible — Saugus clears this bar despite lagging peers on both.

What Overachievers Do Differently: MCAS Grade 10 (ELA)

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 10 shown

Feature	Saugus	Lenox	North Ada	Springfie	Imp
chronic_absenteeism_pct	31.2	18.3	30.4	32.4	+0.622
nss_per_eqv	0.07	0.06	0.31	0.30	+0.442
spend_vs_required_nss	1.03	1.54	1.45	1.26	+0.374
teacher_share_of_spend	0.35	0.37	0.30	0.29	+0.361
nss_per_pupil	17,369	23,945	26,299	26,616	+0.357
teachers_per_lowincome	0.13	0.41	0.12	0.09	+0.352
res_tax_rate	10.7	9.07	17.1	16.1	+0.349
teachers_per_100_student	5.87	10.1	8.67	7.37	+0.346
health_ins_per_capita	588.1	890.8	521.6	534.9	+0.340
ed_budget_share	32.3	48.6	40.3	57.8	-0.163

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	47.0	34.9	+12.1pp
Lenox	90.0	67.3	+22.7pp
North Adams	38.0	15.6	+22.4pp
Springfield	31.0	11.7	+19.3pp
Lee	78.0	60.2	+17.8pp
Quincy	60.0	42.3	+17.7pp
Mashpee	69.0	51.9	+17.1pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Education Budget Share

Predicted 40.9pp · Actual 32.3pp · Gap -8.6pp (higher is better) · Saugus ranks 214 of 219 MA districts on this measure

The Bottom Line

Saugus's education budget share is 32.3%, 8.6pp worse than its demographic prediction of 40.9% — ranking 214 of 219 MA districts, in the bottom 10%. Unlike the academic measures, this gap reflects a municipal budgeting choice, not a demographic constraint.

Top Actionable Levers (what Saugus can change)

Highest-importance actionable factors, Saugus vs. statewide median.

Factor	Saugus	MA median	Importance
Pension & benefits share of budget	25.3%	17.5%	+0.39
School spending ÷ property wealth/capita	0.068	0.082	+0.27
Teacher pay share of school spending	0.349	0.368	+0.18
Residential tax rate (per \$1,000)	10.7	12.6	+0.15
Teachers per 100 ÷ low-income %	0.126	0.277	+0.13

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Pension & benefits share of budget	25.3%	16.1%	-9.2pp
School spending ÷ property wealth/cap	0.068	0.137	+0.069
Teacher pay share of school spending	0.349	0.315	-0.035
Residential tax rate (per \$1,000)	10.7	12.5	+1.87
Teachers per 100 ÷ low-income %	0.126	0.165	+0.039

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

Saugus does worse here than its demographic profile predicts. The strongest RBP driver of Saugus's predicted peer group is pension & benefits share of budget (25.3% vs the 17.5% state median); the widest gap to the over-performing districts is pension & benefits share of budget (-9.2pp, peer median – Saugus). (Driver ranking is close enough at the top that the #1 factor is not fully seed-stable — read the top group, not the exact order.) The top drivers are competing budget categories that crowd out the education share — the lever here sits with Town Meeting and municipal budgeting, not the school district.

What Overachievers Do Differently: Education Budget Share

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 10 shown

Feature	Saugus	Hancock	Springfie	Erving	Imp
fixed_costs_pct	25.3	2.43	19.7	16.3	+0.393
nss_per_eqv	0.07	0.04	0.30	0.06	+0.270
teacher_share_of_spend	0.35	0.38	0.29	0.29	+0.175
res_tax_rate	10.7	2.79	16.1	9.22	+0.149
teachers_per_lowincome	0.13	0.20	0.09	0.21	+0.127
health_ins_per_capita	588.1	11.3	534.9	894.5	+0.090
teachers_per_100_student	5.87	7.36	7.37	9.33	+0.079
spend_vs_required_nss	1.03	1.42	1.26	2.28	+0.064
nss_per_pupil	17,369	21,836	26,616	34,568	+0.047
chronic_absenteeism_pct	31.2	16.9	32.4	18.9	-0.026

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	32.3	40.9	-8.6pp
Hancock	78.0	52.1	+25.9pp
Springfield	57.8	42.1	+15.7pp
Erving	45.1	32.5	+12.6pp
Sunderland	57.1	44.9	+12.2pp
Chelsea	52.2	40.4	+11.9pp
Truro	24.6	14.4	+10.1pp

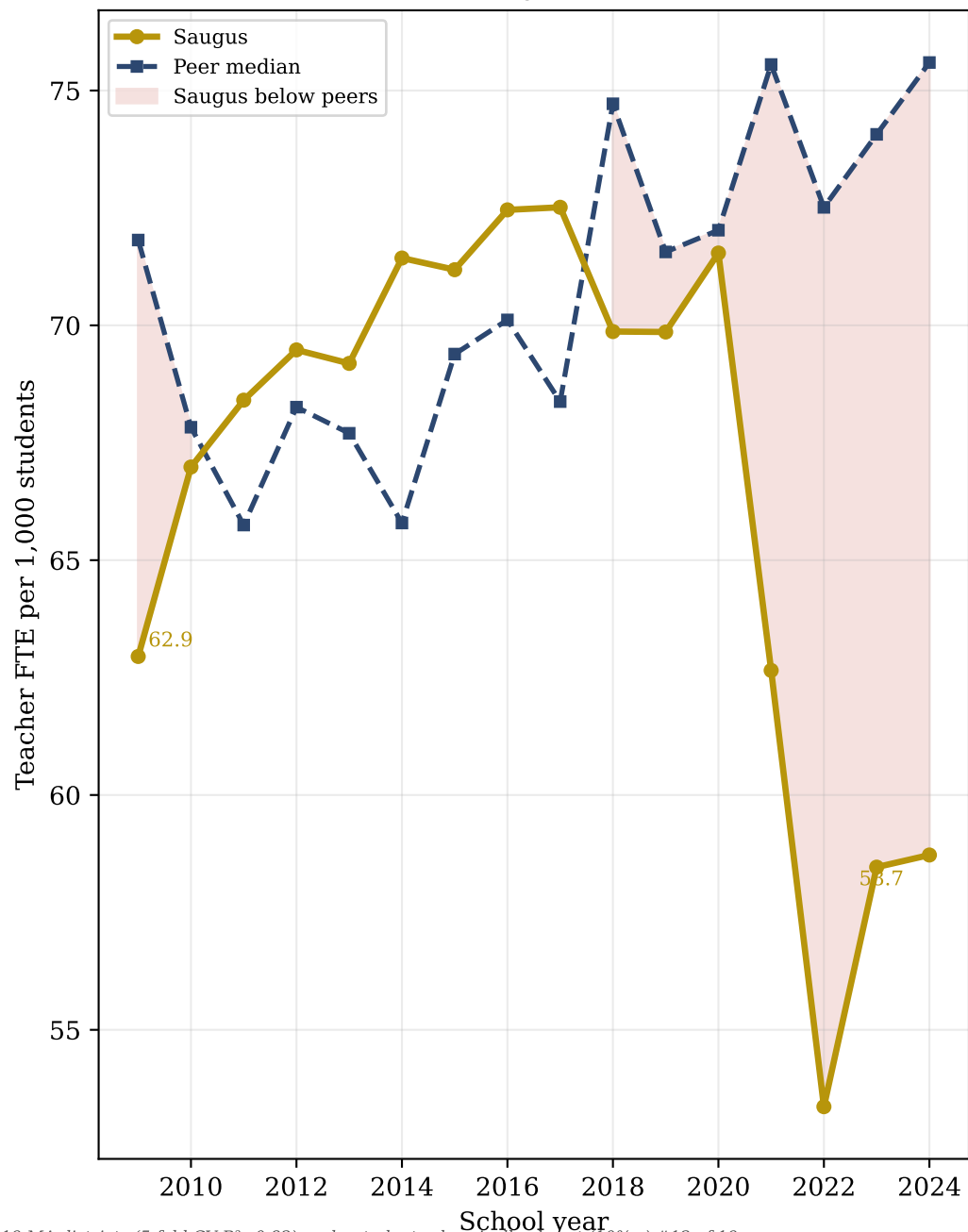
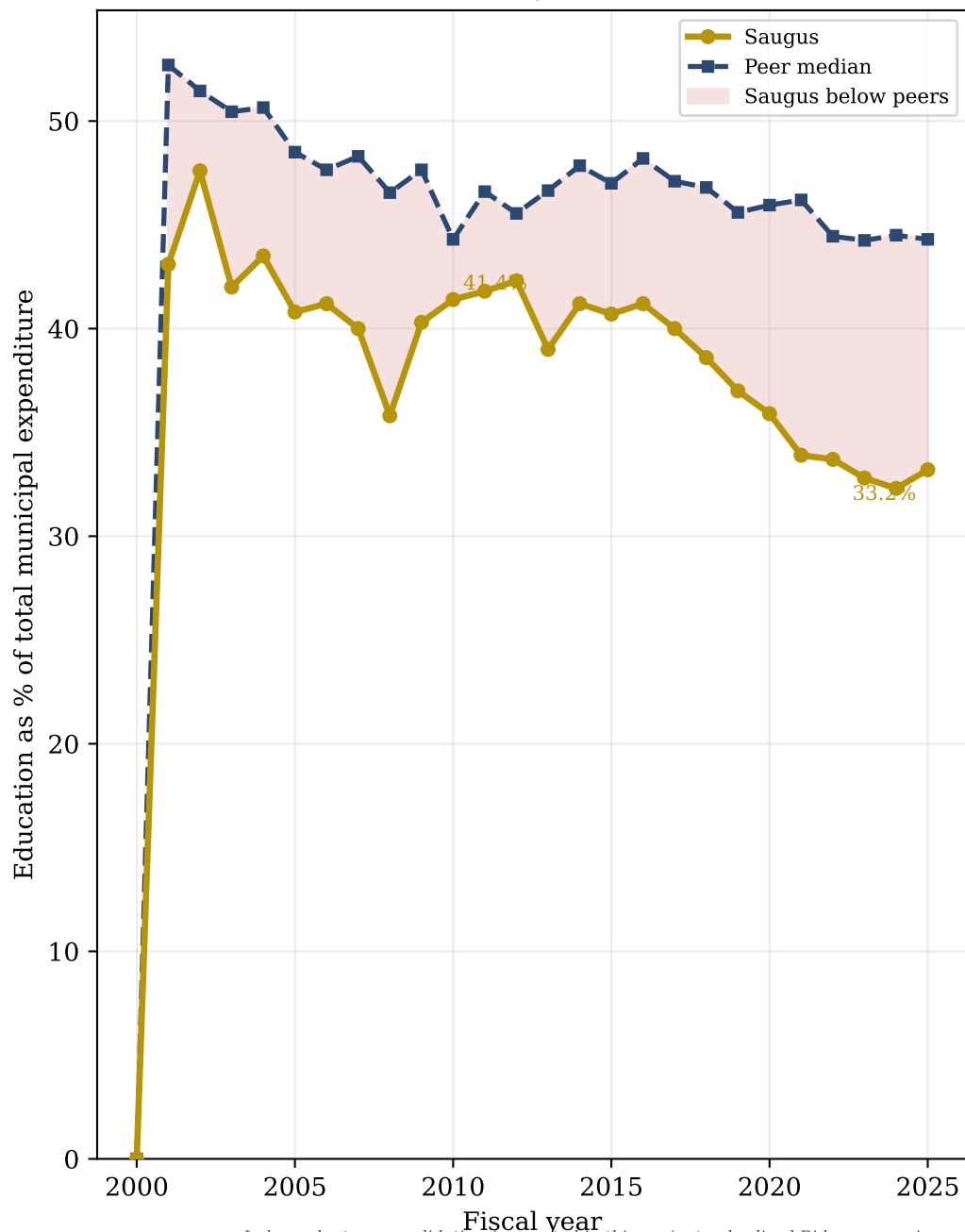
Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

Budget Allocation & Teacher Density — 15-Year Trajectory

Source: MA DLS Schedule A (budget) · MA DESE Staffing (teachers) · Peer comparison: 8 demographically similar overachiever towns

Education Budget Share
(Schedule A, general fund)

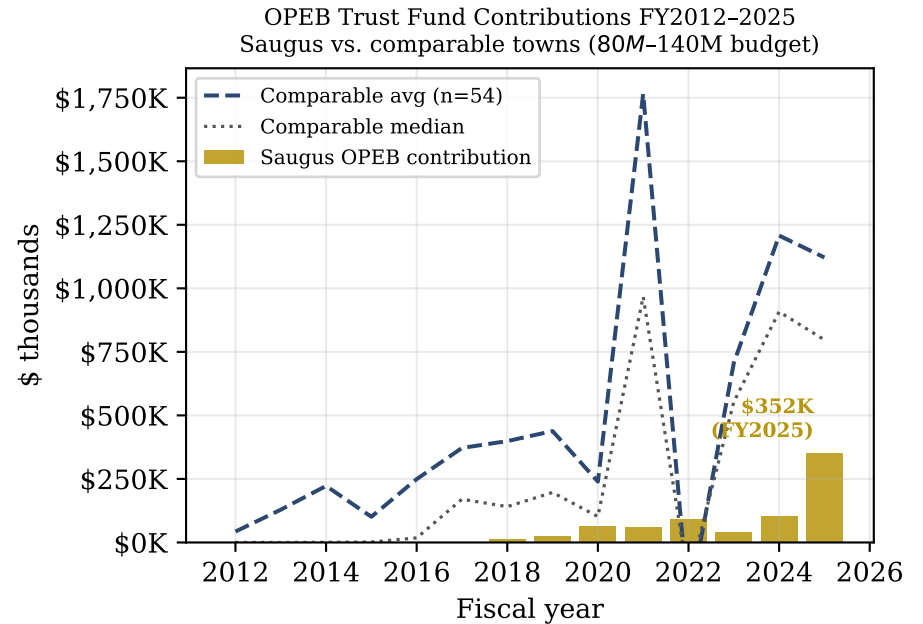
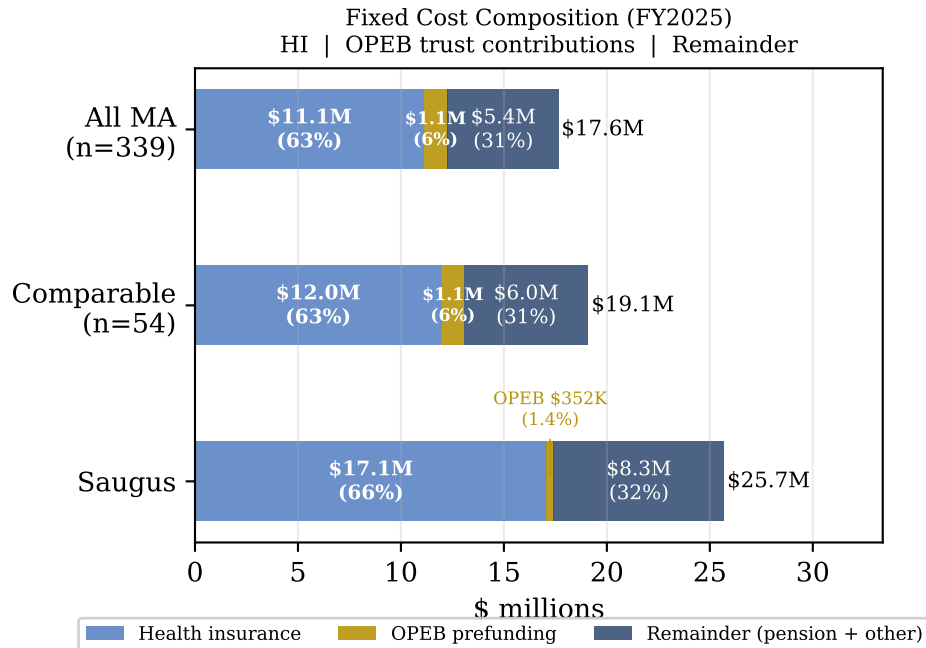
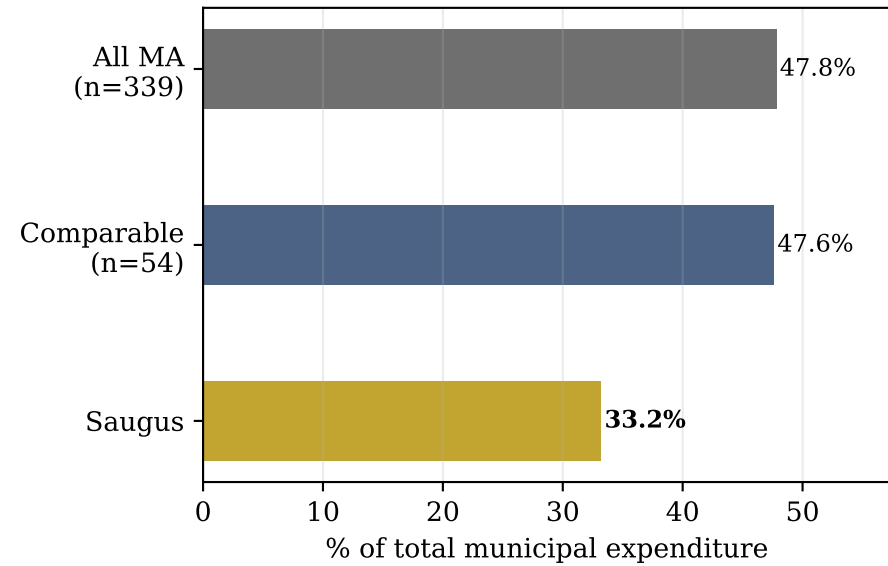
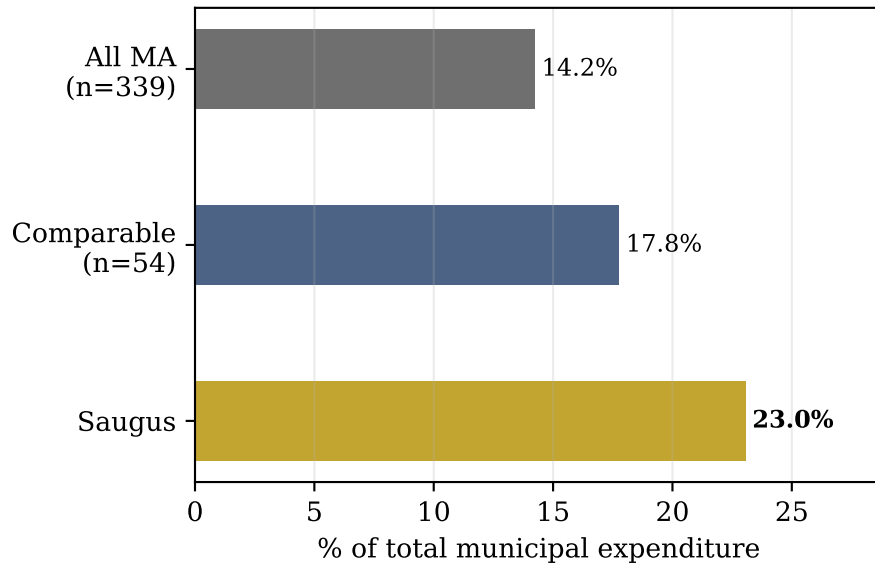
Teacher Density
(DESE Staffing + Enrollment)



Independent cross-validation (computed in this run): standardized Ridge regression on 219 MA districts (5-fold CV $R^2=0.82$) ranks students chronically absent (10%+) #12 of 19 standardized predictors of MCAS 3–8 ($|standardized \beta|=0.08$) — a divergence, not corroboration: a linear model assigns students chronically absent (10%+) a small standardized coefficient, whereas RBP's Exhibit 5 ranks it among the top drivers — the kind of nonlinear / interaction effect RBP is designed to surface and linear regression misses. Peer set: 8 demographically similar overachiever towns (Mahalanobis $d_M \leq 2$).

Fixed Costs: Sources, Scale, and OPEB Prefunding

FY2025 · Comparable = 54 MA towns (80M-140M) · All MA = 339 · Columns sum exactly to total for all 4,898 statewide municipality-years



Optimum Profile — What Overachievers Look Like

Each grid shows actionable feature values for Saugus vs overachiever towns. Showing both peer groups tests whether the finding holds across definitions.

Overachievers — all 20 MA districts that beat their demographic prediction

Feature	Saugus	Springfie	North Ada	Fitchburg	Quincy	Amherst	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	32.4%	30.4%	34.0%	16.5%	16.7%	19.9%	-11.3pp	20
Teachers / 100 students	5.9	7.4	8.7	7.6	7.5	9.8	8.3	+2.4	20
Avg teacher salary	\$89,778	\$84,692	\$76,435	\$84,723	\$94,955	\$87,842	\$84,708	\$-5,070	20
Net school spending / pupil	\$17,369	\$26,616	\$26,299	\$20,866	\$20,257	\$31,268	\$22,700	+\$+5,330	20
Shown: Springfield, North Adams, Fitchburg, Quincy, Amherst (+ 15 more in median)									

Demographically Similar Overachievers — 17 towns within 2σ Mahalanobis distance of Saugus on income, property wealth, poverty, size, education & ELL

Feature	Saugus	Norwood	Clinton	Westport	Lenox	Acushnet	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	25.1%	7.7%	13.8%	18.3%	9.2%	18.3%	-12.9pp	17
Teachers / 100 students	5.9	8.8	7.3	7.5	10.1	6.1	8.4	+2.5	17
Avg teacher salary	\$89,778	\$92,324	\$87,059	\$83,017	\$84,669	\$77,654	\$84,723	\$-5,055	17
Net school spending / pupil	\$17,369	\$21,084	\$18,512	\$18,099	\$23,945	\$16,414	\$21,084	+\$+3,715	17
Shown: Norwood, Clinton, Westport, Lenox, Acushnet (+ 12 more in median)									

Similarity metric: Mahalanobis distance $d_M = \sqrt{(\Delta x \cdot \Sigma^{-1} \cdot \Delta x')}$, Σ = sample covariance of all MA districts on (median_hh_income, equalized_income, low_income_pct, total_enrollment, pct_bachelors_plus, ell_pct). Threshold: $d_M \leq 2$ (2σ). Pre-specified; not tuned to result. Gap = peer median – Saugus. Red = Saugus below peer median, Green = at or above.